

# Natural Language Processing

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Learning Outcome 3

# Objective

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After attending this session, you should be able

- Introduction to NLP and applications
- Installation of packages
- Tokenizing the text data
- Stemming
- Lemmatization
- Dividing text data into chunks
- Extracting frequency of terms using BoW model
- NLP pipeline
- Practical implementation of Ecommerce item classification



# NLP pipeline

## Sentence Tokenization

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Dr. Himanshu ordered samosas, dosa, panipuri etc. for his breakfast

- Sentence 1 = Dr
- Sentence 2 = Himanshu ordered samosas, dosa, panipuri etc
- Sentence 3 = for his breakfast
- Sentence tokenizer (splitter) (NLTK, spacy has ready to used tokenizer)

# Word Tokenization

"So here's another post complaining about..."

Word Tokenization

So

here

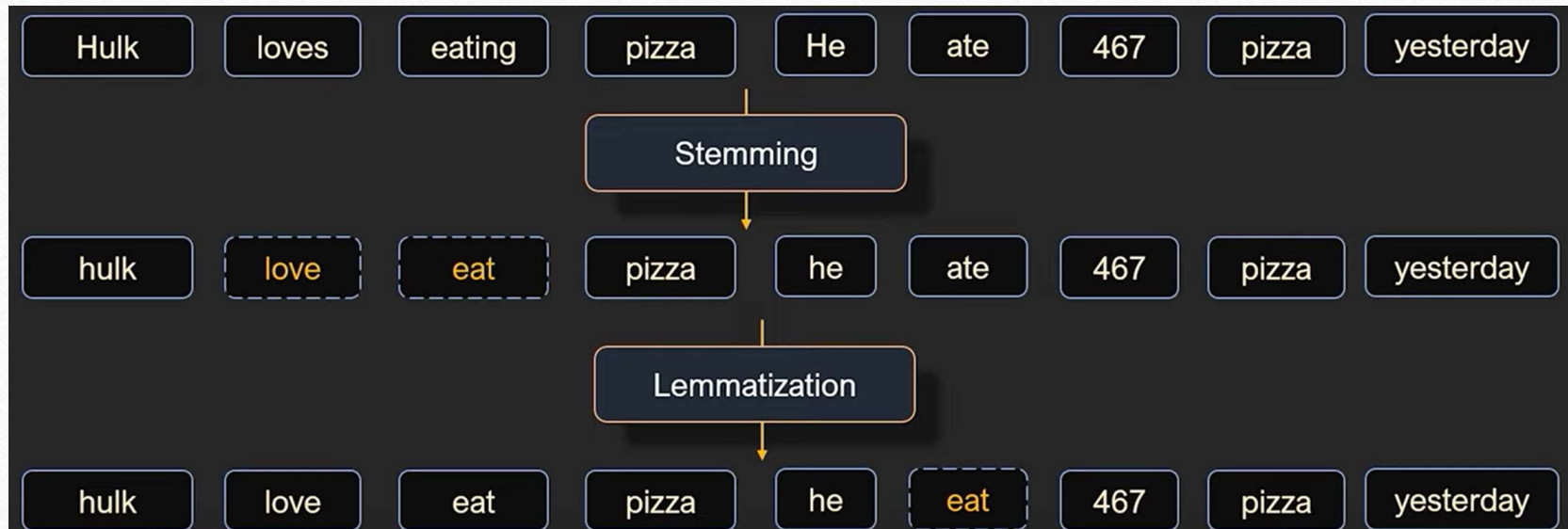
's

another

post



# Stemming and Lemmatization



## Stemming

adjustable -> adjust  
formality -> formaliti  
formaliti -> formal  
airliner -> airlin

## Lemmatization

was -> (to) be  
better -> good  
meeting -> meeting

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- Pre-processing

Sentence Tokenization



Word Tokenization



Stemming, Lemmatization

## NLP Pipeline

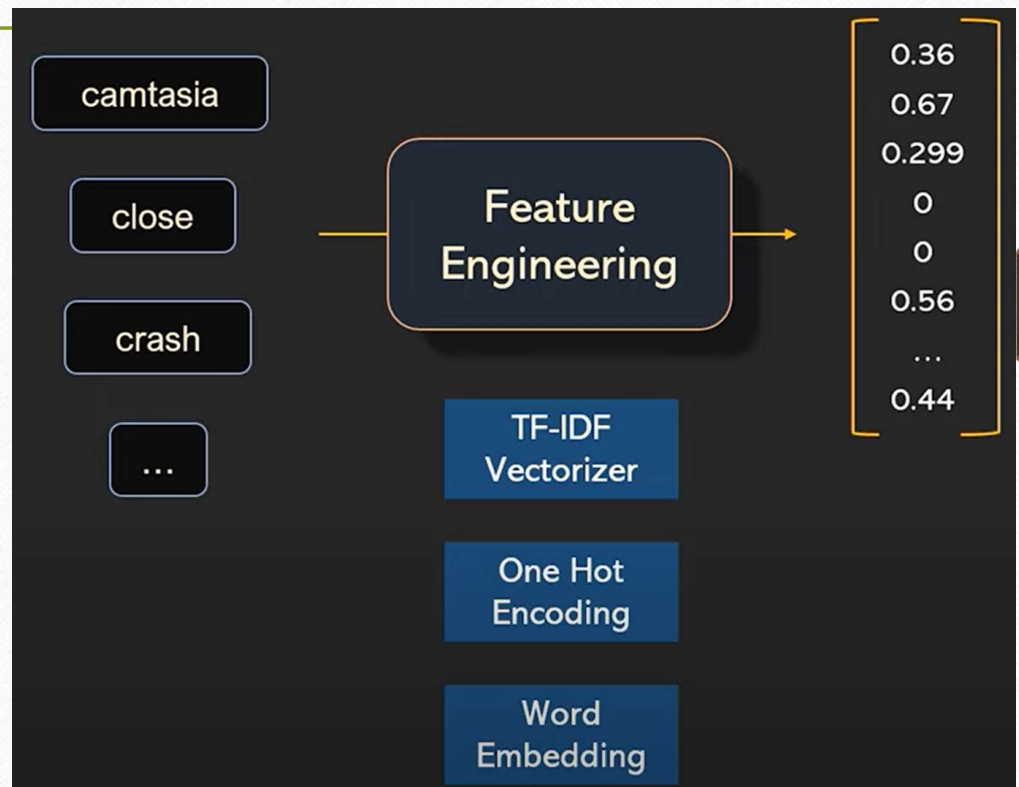
Data  
Acquisition

Text Extraction  
& Cleanup

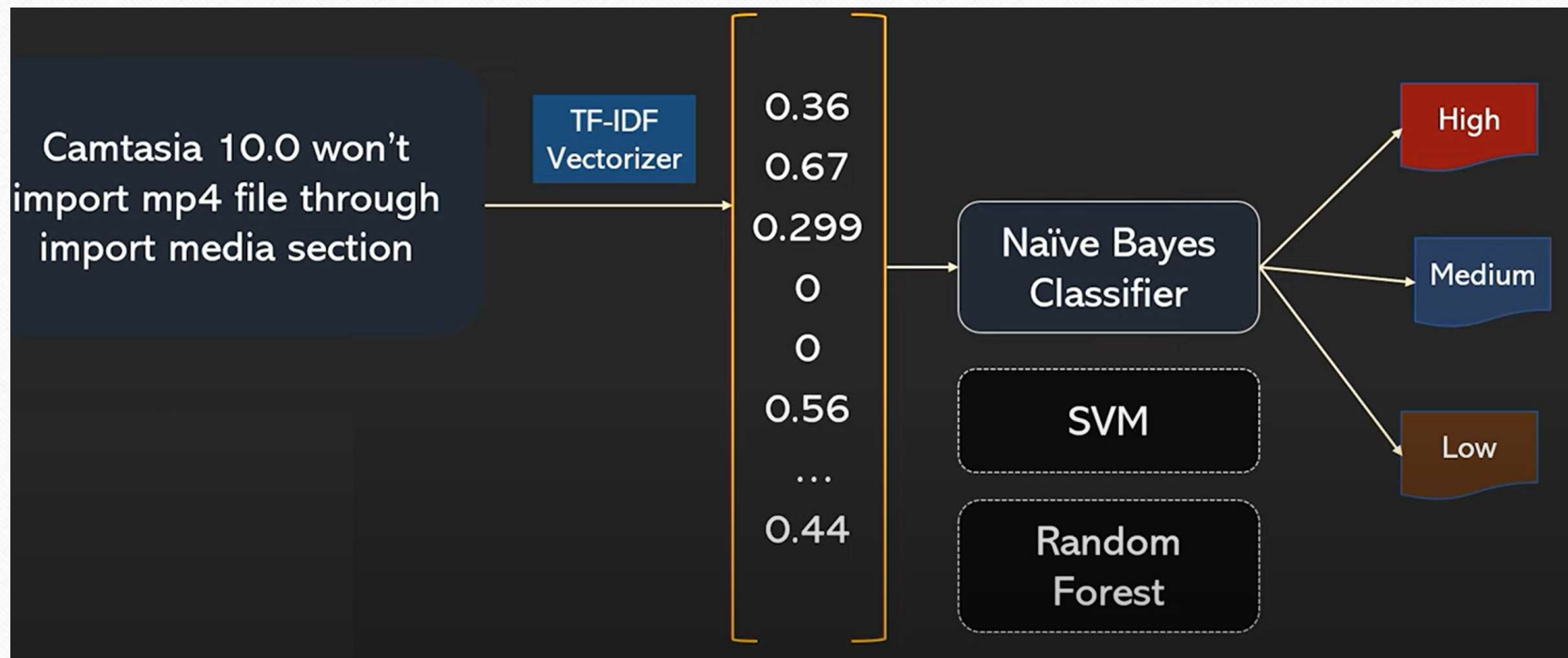
Pre-processing



# Feature Engineering

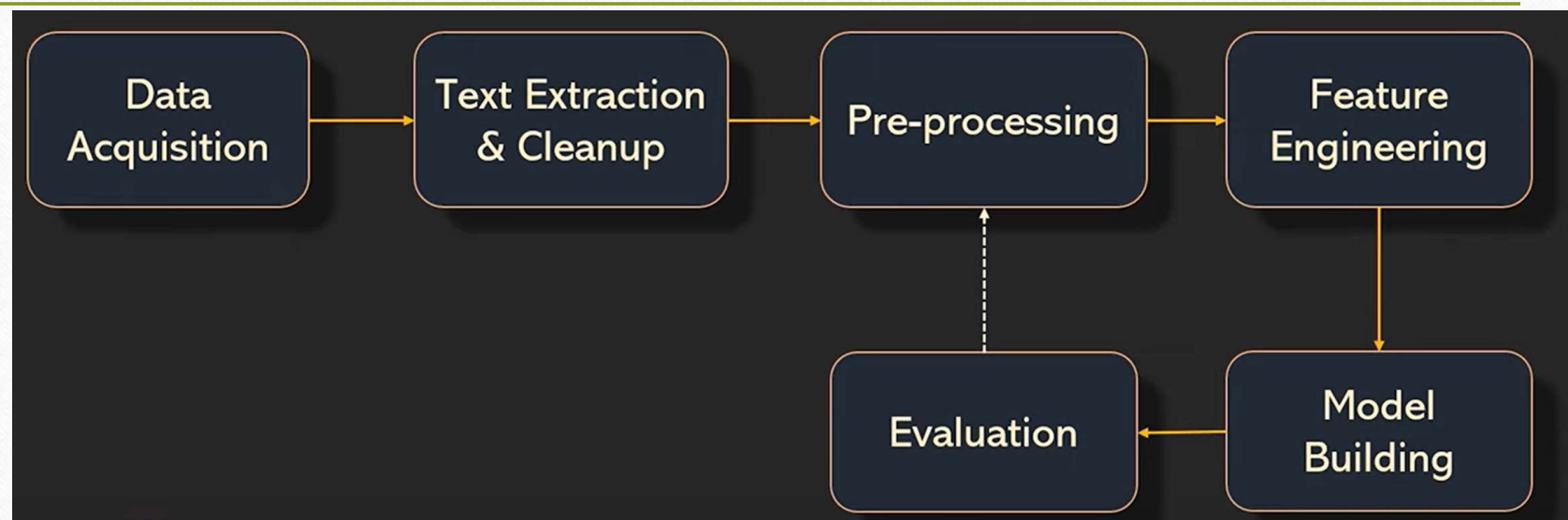


# Customer complain severity classification





# NLP pipeline



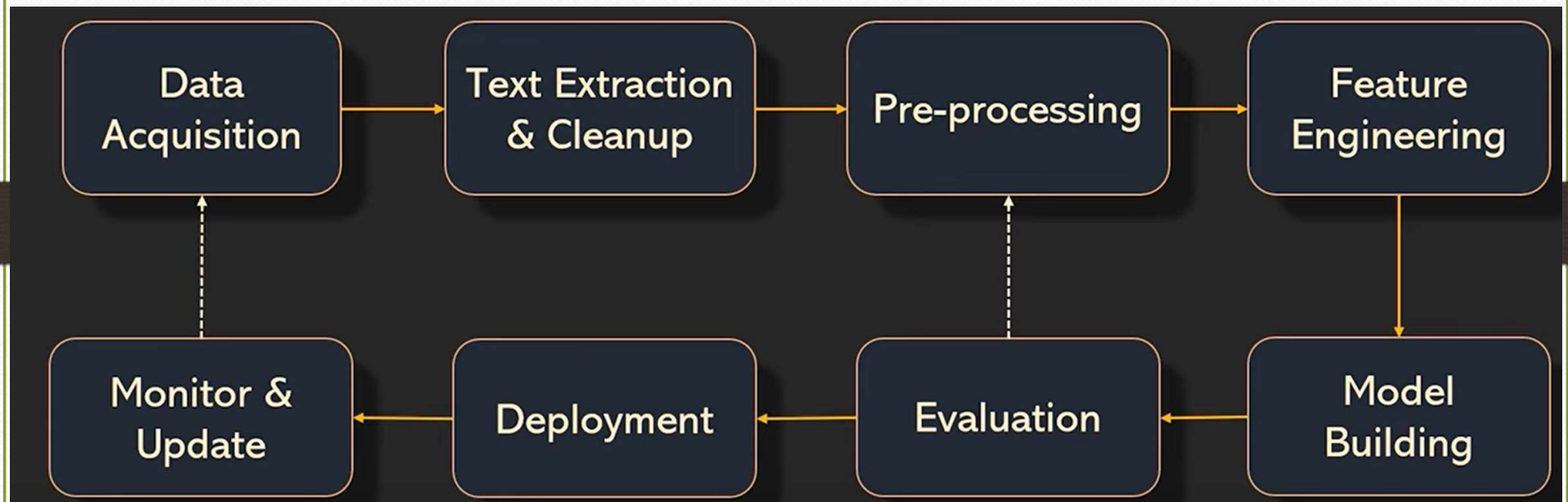
# End to end machine learning platform

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**H2O.ai**





# Exercise

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Time: 30 minutes

Presentation time per group: 10 minutes (use ppt, whiteboard, practical demo)

- Group 1: Definition of NLP and its real-world applications
- Group 2: Various techniques used in NLP
- Group 3: Label Encoding and One-Hot Encoding for text representation
- Group 4: Bag of Words (BoW) for text representation
- Group 5: SpaCy and NLTK libraries
- Group 6: Word Embeddings (Word2Vec)



# NLP pipeline: classwork

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Refer NLP book reference and perform following tasks:

- Installing NLTK, genism, pattern
- Word, sentence tokenization
- stemming and lemmatization
- Chunking
- Bag of Words representation

# Bag of Word representation

- **Review 1:** This movie is very scary and long
- **Review 2:** This movie is not scary and is slow
- **Review 3:** This movie is spooky and good

|             | 1<br>This | 2<br>movie | 3<br>is | 4<br>very | 5<br>scary | 6<br>and | 7<br>long | 8<br>not | 9<br>slow | 10<br>spooky | 11<br>good | Length of the<br>review(in<br>words) |
|-------------|-----------|------------|---------|-----------|------------|----------|-----------|----------|-----------|--------------|------------|--------------------------------------|
| Review<br>1 | 1         | 1          | 1       | 1         | 1          | 1        | 1         | 0        | 0         | 0            | 0          | 7                                    |
| Review<br>2 | 1         | 1          | 2       | 0         | 0          | 1        | 1         | 0        | 1         | 0            | 0          | 8                                    |
| Review<br>3 | 1         | 1          | 1       | 0         | 0          | 0        | 1         | 0        | 0         | 1            | 1          | 6                                    |



# Limitation of BoW

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- If the new sentences contain new words, then our vocabulary size would increase and thereby, the length of the vectors would increase too.
- Additionally, the vectors would also contain many 0s, thereby resulting in a sparse matrix (which is what we would like to avoid)
- We are retaining no information on the grammar of the sentences nor on the ordering of the words in the text.

# Text Representation using TF-IDF

$$(tf\_idf)_{t,d} = tf_{t,d} * idf_t \quad tf_{t,d} = \frac{n_{t,d}}{\text{Number of terms in the document}}$$

- *Review 2: This movie is not scary and is slow*

$$TF('movie') = 1/8$$

$$TF('long') = 0/8 = 0$$

$$TF('is') = 2/8 = 1/4$$

$$TF('not') = 1/8$$

$$TF('very') = 0/8 = 0$$

$$TF('slow') = 1/8$$

$$TF('scary') = 1/8$$

$$TF('spooky') = 0/8 = 0$$

$$TF('and') = 1/8$$

$$TF('good') = 0/8 = 0$$



# Inverse Document Frequency (IDF)

$$idf_t = \log \frac{\text{number of documents}}{\text{number of documents with term 't'}}$$

$$\text{IDF}(\text{'movie', }) = \log(3/3) = 0$$

$$\text{IDF}(\text{'is'}) = \log(3/3) = 0$$

$$\text{IDF}(\text{'not'}) = \log(3/1) = \log(3) = 0.48$$

$$\text{IDF}(\text{'scary'}) = \log(3/2) = 0.18$$

$$\text{IDF}(\text{'and'}) = \log(3/3) = 0$$

$$\text{IDF}(\text{'slow'}) = \log(3/1) = 0.48$$

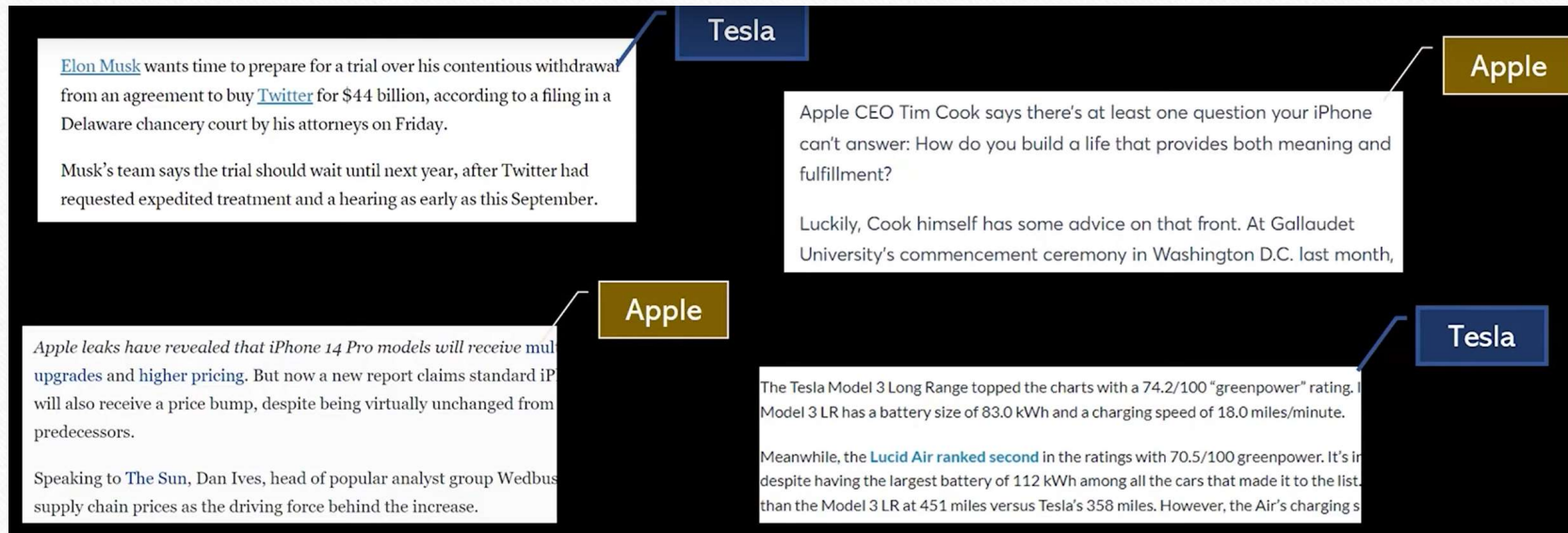
# TF-IDF Representation

| Term   | Review 1 | Review 2 | Review 3 | TF (Review 1) | TF (Review 2) | TF (Review 3) | IDF  | TF-IDF (Review 1) | TF-IDF (Review 2) | TF-IDF (Review 3) |
|--------|----------|----------|----------|---------------|---------------|---------------|------|-------------------|-------------------|-------------------|
| This   | 1        | 1        | 1        | 1/7           | 1/8           | 1/6           | 0.00 | 0.000             | 0.000             | 0.000             |
| movie  | 1        | 1        | 1        | 1/7           | 1/8           | 1/6           | 0.00 | 0.000             | 0.000             | 0.000             |
| is     | 1        | 2        | 1        | 1/7           | 1/4           | 1/6           | 0.00 | 0.000             | 0.000             | 0.000             |
| very   | 1        | 0        | 0        | 1/7           | 0             | 0             | 0.48 | 0.068             | 0.000             | 0.000             |
| scary  | 1        | 1        | 0        | 1/7           | 1/8           | 0             | 0.18 | 0.025             | 0.022             | 0.000             |
| and    | 1        | 1        | 1        | 1/7           | 1/8           | 1/6           | 0.00 | 0.000             | 0.000             | 0.000             |
| long   | 1        | 0        | 0        | 1/7           | 0             | 0             | 0.48 | 0.068             | 0.000             | 0.000             |
| not    | 0        | 1        | 0        | 0             | 1/8           | 0             | 0.48 | 0.000             | 0.060             | 0.000             |
| slow   | 0        | 1        | 0        | 0             | 1/8           | 0             | 0.48 | 0.000             | 0.060             | 0.000             |
| spooky | 0        | 0        | 1        | 0             | 0             | 1/6           | 0.48 | 0.000             | 0.000             | 0.080             |
| good   | 0        | 0        | 1        | 0             | 0             | 1/6           | 0.48 | 0.000             | 0.000             | 0.080             |



# Text classification

## News article classification problem



# Bag of Word Representation

|                    | musk | that | price | market | investor | iphone | itunes | gigafactory | ...   |
|--------------------|------|------|-------|--------|----------|--------|--------|-------------|-------|
| Apple<br>article 1 | [ 0  | 32   | 45    | 48     | 26       | 7      | 3      | 0           | ... ] |
| Apple<br>article 2 | [ 0  | 4    | 3     | 7      | 0        | 6      | 3      | 0           | ... ] |
| Tesla<br>article 3 | [ 15 | 31   | 44    | 43     | 25       | 0      | 0      | 0           | ... ] |
| Tesla<br>article 4 | [ 3  | 0    | 0     | 0      | 0        | 0      | 0      | 1           | ... ] |



# Bag of Word Representation

|           | musk | that | price | market | investor | iphone | itunes | gigafactory | ...   |
|-----------|------|------|-------|--------|----------|--------|--------|-------------|-------|
| Apple     |      |      |       |        |          |        |        |             |       |
| article 1 | [ 0  | 32   | 45    | 48     | 26       | 7      | 3      | 0           | ... ] |
| Apple     |      |      |       |        |          |        |        |             |       |
| article 2 | [ 0  | 4    | 3     | 7      | 0        | 6      | 3      | 0           | ... ] |
| Tesla     |      |      |       |        |          |        |        |             |       |
| article 3 | [ 15 | 31   | 44    | 43     | 25       | 0      | 0      | 0           | ... ] |
| Tesla     |      |      |       |        |          |        |        |             |       |
| article 4 | [ 3  | 0    | 0     | 0      | 0        | 0      | 0      | 1           | ... ] |

that → 3      gigafactory → 1      iphone → 2

**Document Frequency (DF)** = Number of times term *t* is present in all docs

that → 4/3 → 1.33      gigafactory → 4/1 → 4      iphone → 4/2 → 2

$$IDF(t) = \log \left( \frac{\text{Total Documents}}{\text{Number of documents term } t \text{ is present in}} \right)$$

# Why log function?

$$IDF(t) = \log \left( \frac{\text{Total Documents}}{\text{Number of documents term } t \text{ is present in}} \right)$$

Here is the intuition: If term frequency for the word 'computer' in doc1 is 10 and in doc2 it's 20, we can say that doc2 is more relevant than doc1 for the word 'computer'.

However, if the term frequency of the same word, 'computer', for doc1 is 1 million and doc2 is 2 millions, at this point, there is no much difference in terms of relevancy anymore because they both contain a very high count for term 'computer'.

Just like Debasis's answer, adding log is to dampen the importance of term that has a high frequency, e.g. Using log base 2, the count of 1 million will be reduced to 19.9!



# Sklearn IDF function

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If `smooth_idf=True` (the default), the constant "1" is added to the numerator and denominator of the idf as if an extra document was seen containing every term in the collection exactly once, which prevents zero divisions:  $\text{idf}(t) = \log \left[ \frac{1 + n}{1 + \text{df}(t)} \right] + 1$ .

# Exercise: Ecommerce item classification

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- Problem statement: Train and evaluate the model for ecommerce item classification.



# Summary

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- Introduced NLP and list out applications
- Installed of packages
- Tokenizing the text data
- Stemming
- Lemmatization
- Dividing text data into chunks
- Extracting frequency of terms using BoW model
- NLP pipeline
- Practical implementation of Ecommerce item classification



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